



US 20250268380A1

(19) **United States**

(12) **Patent Application Publication**
Lin et al.

(10) **Pub. No.: US 2025/0268380 A1**

(43) **Pub. Date: Aug. 28, 2025**

(54) **QUICK-DETACHABLE AND FOLDABLE
INFLATABLE SEAT**

(71) Applicant: **Shandong Transe Group Co.,LTD,**
Weihai (CN)

(72) Inventors: **Peng Lin,** Weihai (CN); **Youqiang
Wang,** Weihai (CN); **Xiaojie Gong,**
Weihai (CN)

(21) Appl. No.: **18/586,609**

(22) Filed: **Feb. 26, 2024**

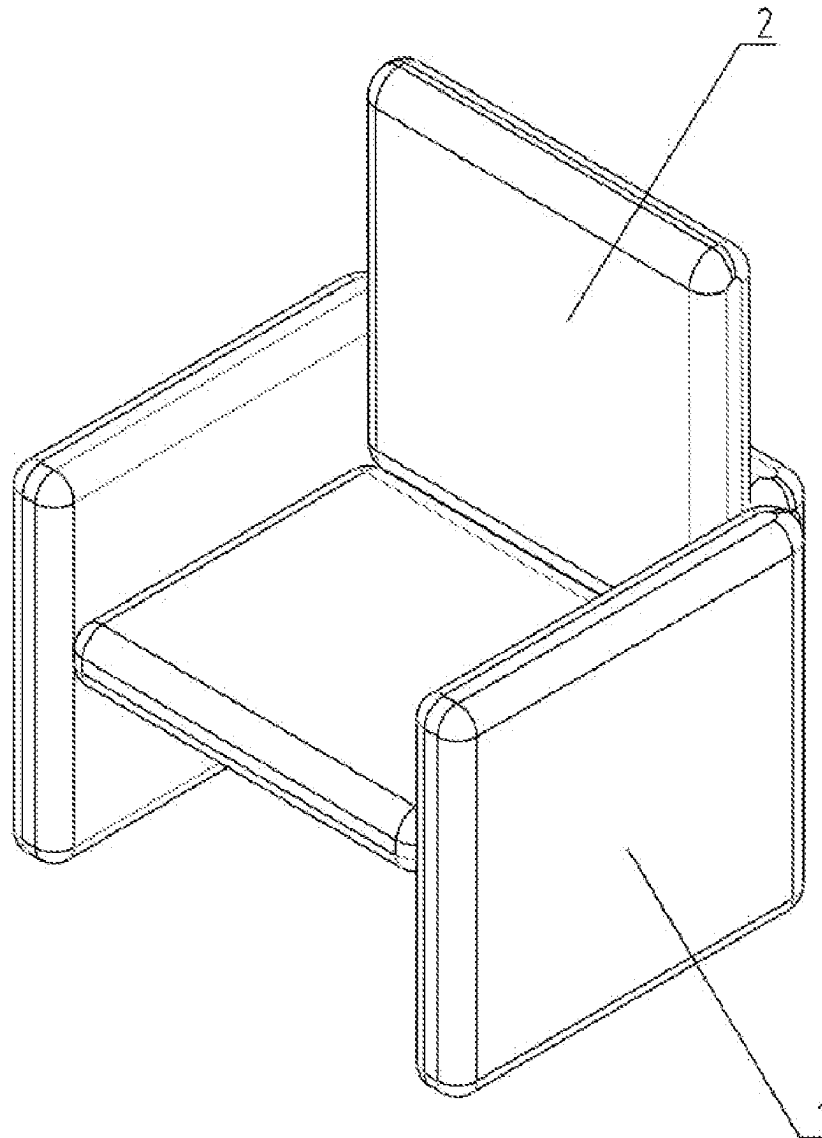
Publication Classification

(51) **Int. Cl.**
A47C 4/54 (2006.01)
A47C 4/28 (2006.01)

(52) **U.S. Cl.**
CPC . **A47C 4/54** (2013.01); **A47C 4/28** (2013.01)

(57) **ABSTRACT**

The present disclosure provides a quick-detachable and foldable inflatable seat, including a first inflatable plate and a second inflatable plate. The first inflatable plate and the second inflatable plate, after being inflated, are assembled to form an inflatable seat by folding. The first inflatable plate is divided into a left support plate, a rear support plate and a right support plate. The second inflatable plate is divided into a back plate and a seating plate, with a crease arranged therebetween. The first inflatable plate or upper and lower portions of the second inflatable plate is connected via sewing threads at the creases, and sealing strips are arranged on upper and lower sides of the sewing threads at the creases. The inflatable seat features simple structure, small size, regular folding, low risk of air leakage and easy disassembly and assembly.



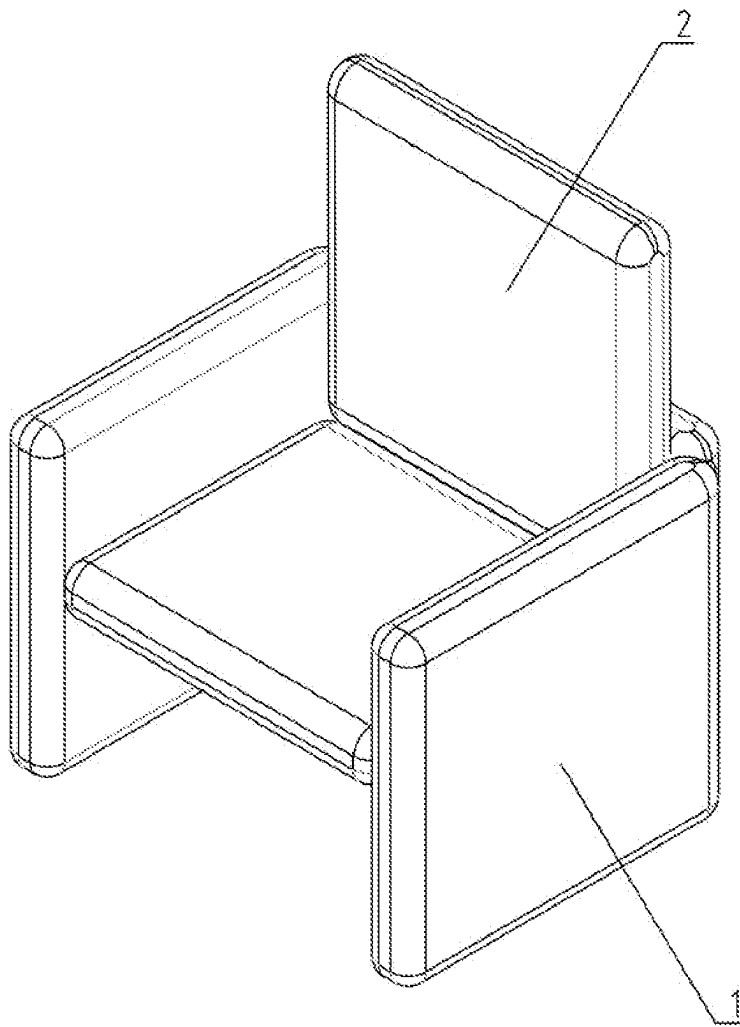


FIG. 1

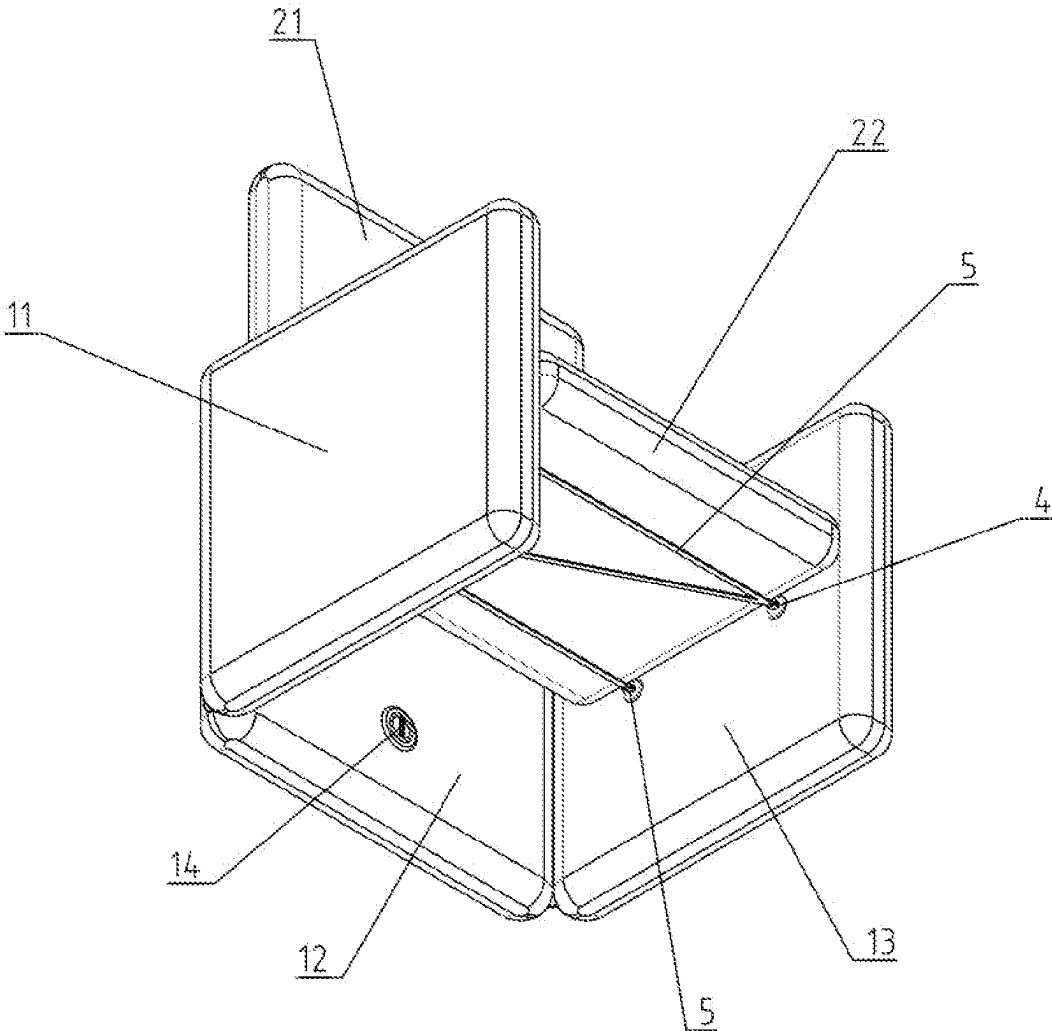


FIG. 2

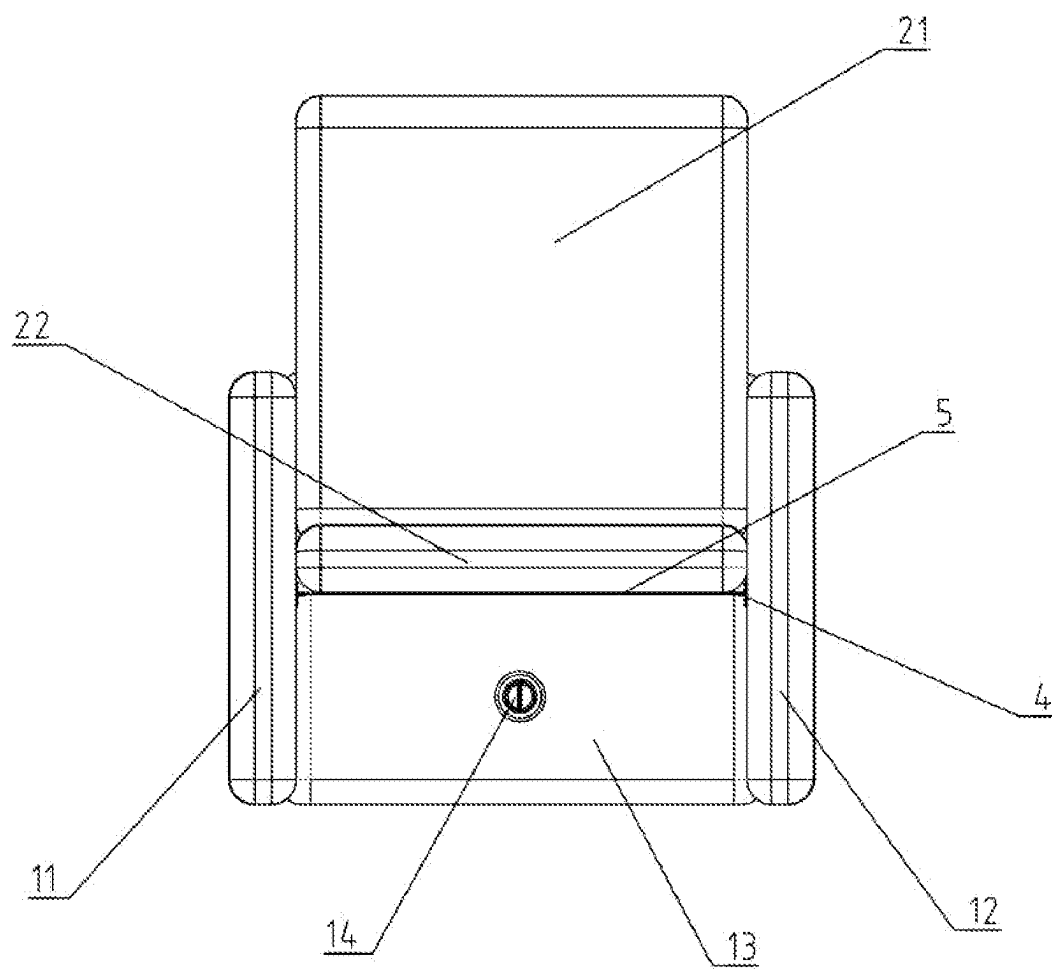


FIG. 3

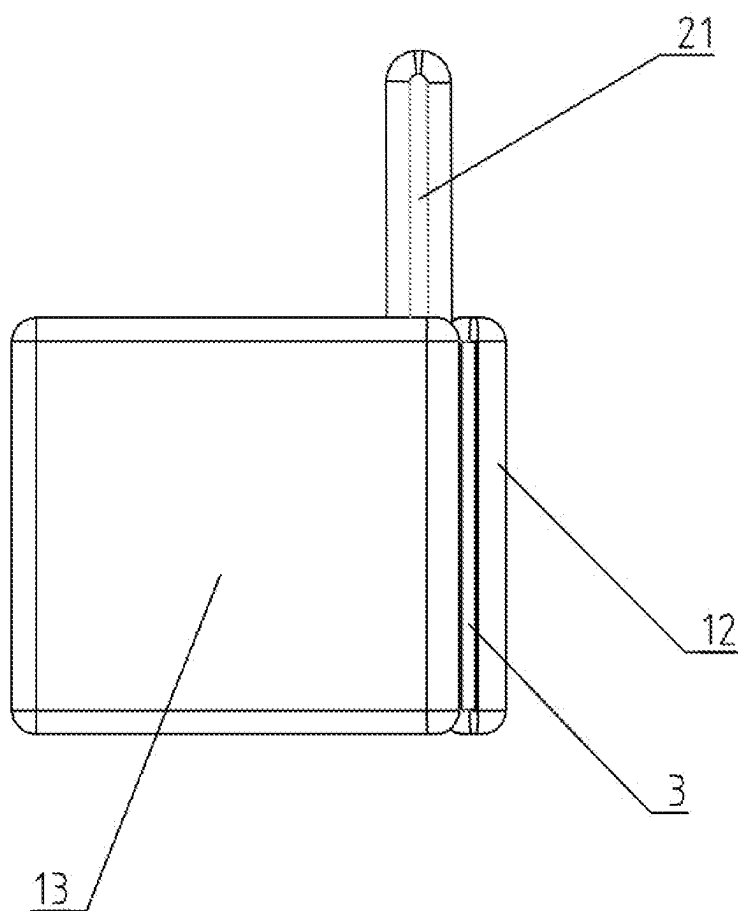


FIG. 4

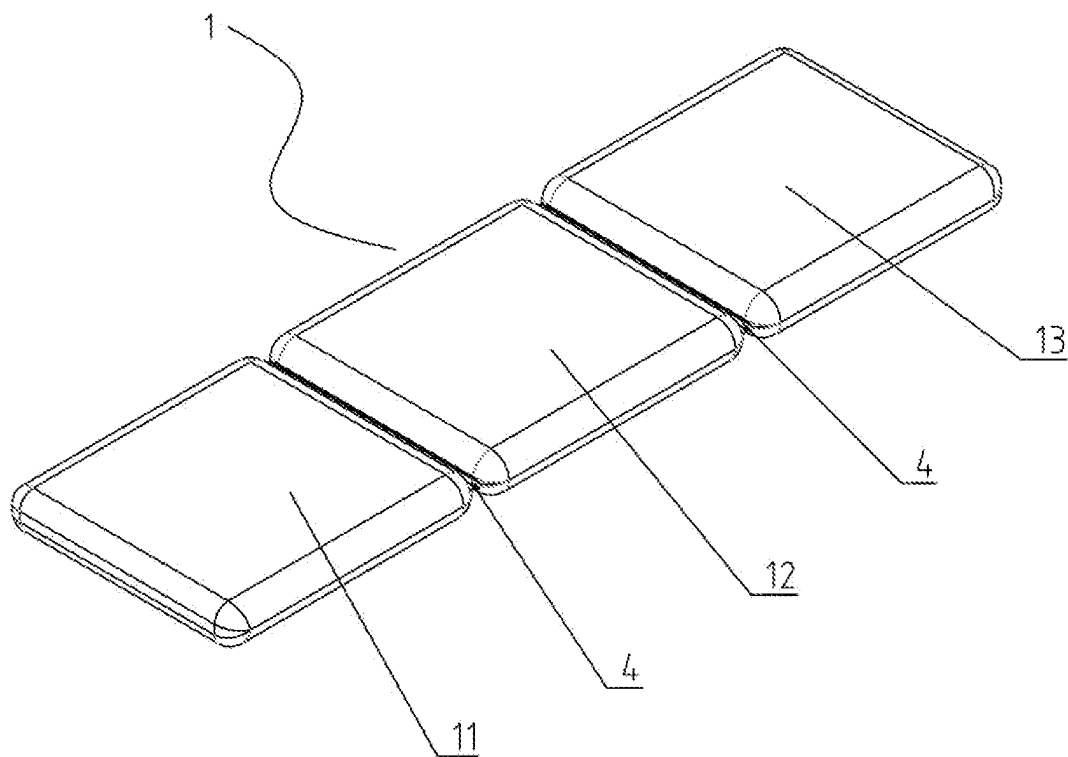


FIG. 5

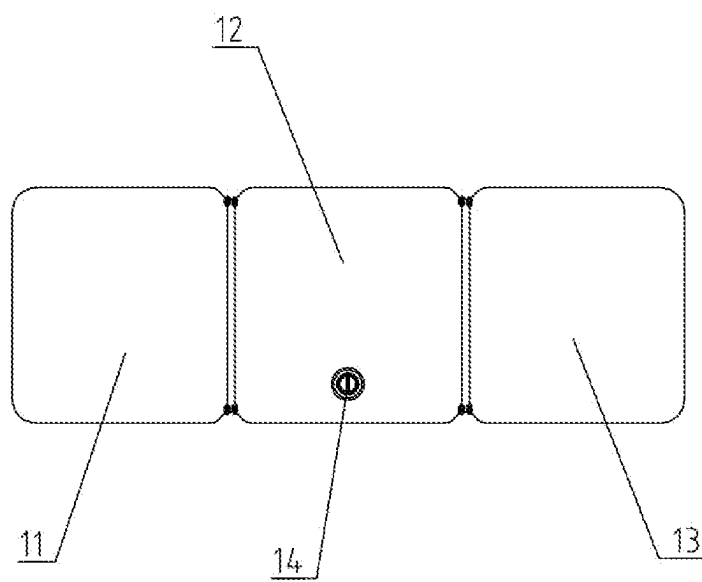


FIG. 6

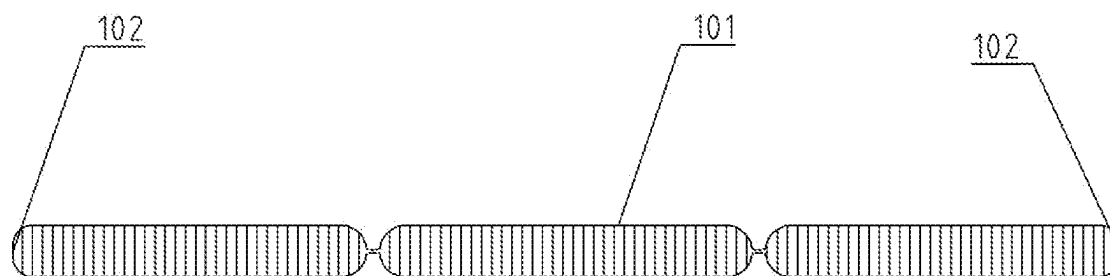


FIG. 7

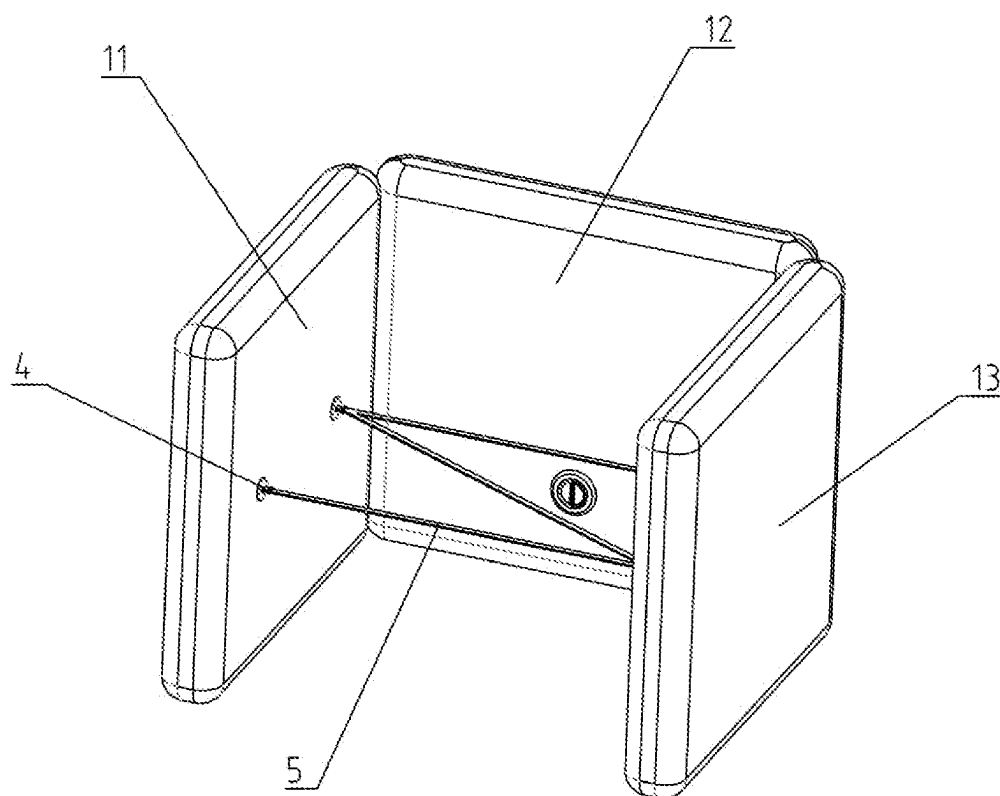


FIG. 8

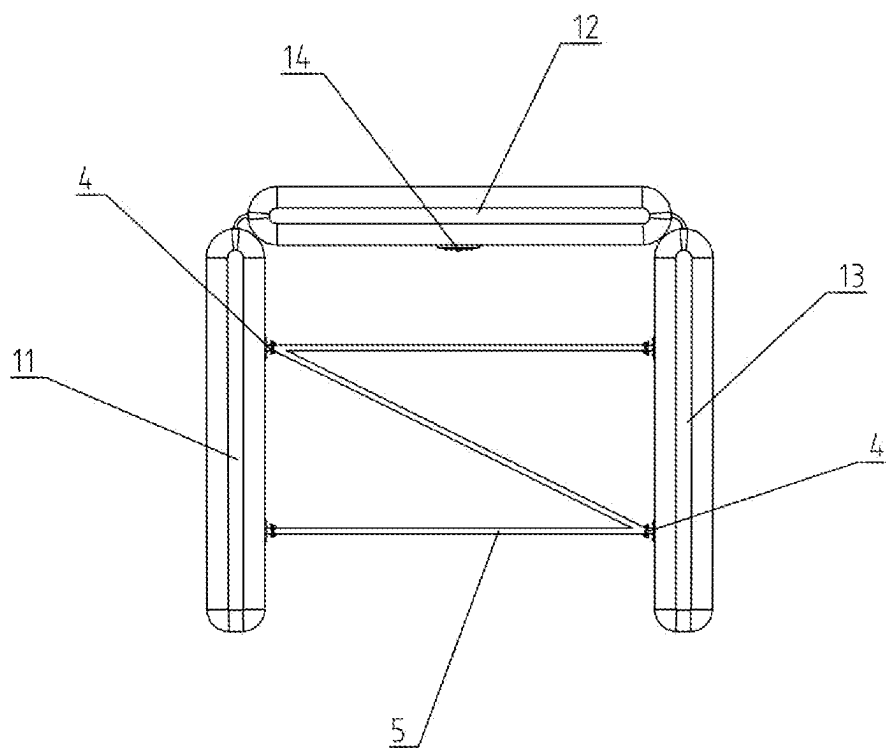


FIG. 9

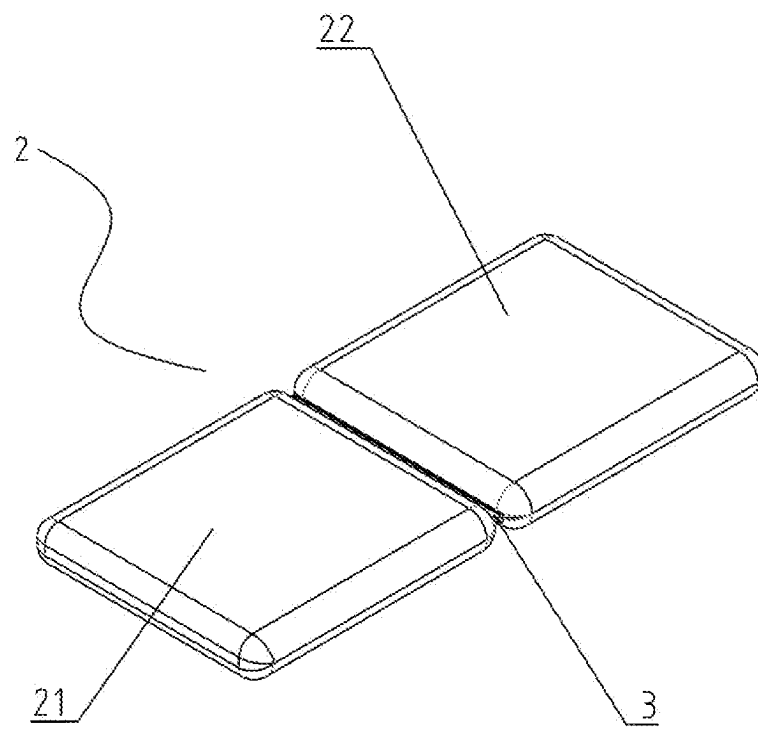


FIG. 10

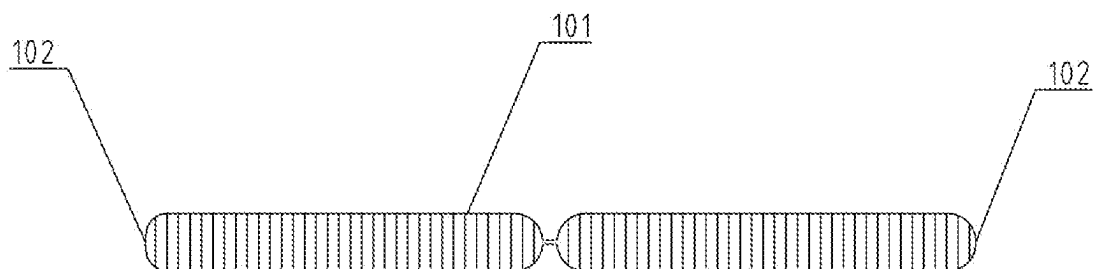


FIG. 11

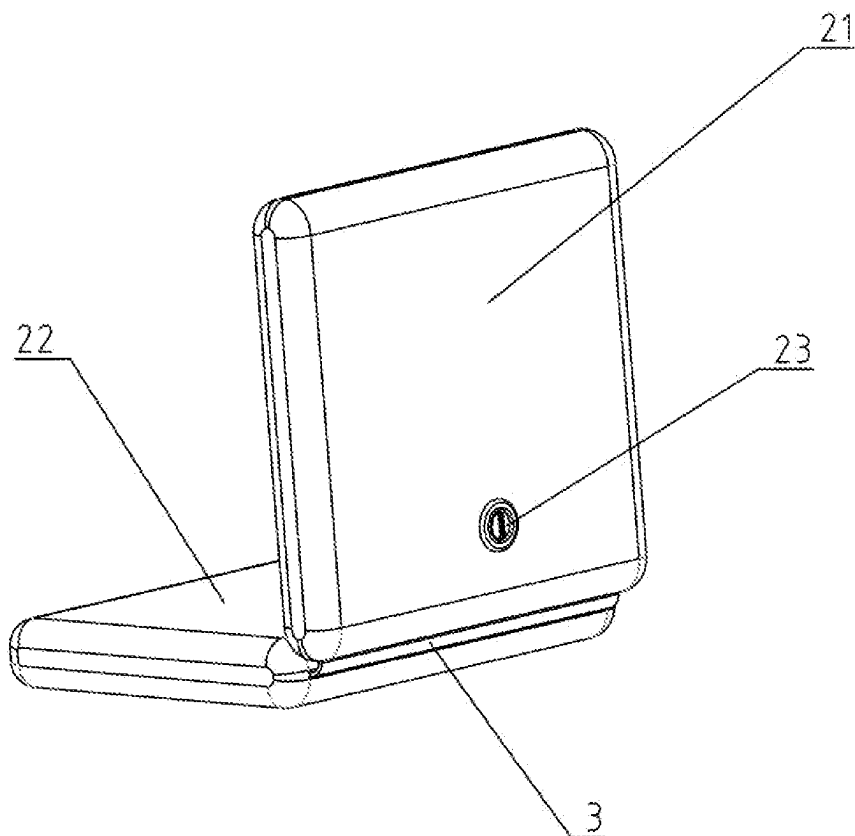


FIG. 12

QUICK-DETACHABLE AND FOLDABLE INFLATABLE SEAT

TECHNICAL FIELD

[0001] The present disclosure relates to the technical field of inflatable seats, and in particular, to a quick-detachable and foldable inflatable seat.

BACKGROUND

[0002] With the improvement of living standards, traveling has been a popular manner for leisure and entertainment. As traditional chairs are bulky, not easy to fold and carry, inflatable seats are invented.

[0003] Most of the existing inflatable seats are of integral and multi-chamber structures, which need to be inflated several times, and there are problems involving high risk of air leakage, and inconvenient folding for storage and secondary utilization. In addition, the existing structure leads to a large packaging volume, and when folded for storage, the seat occupies a lot of space, and cannot be folded neatly, which may damage the wire drawing cloth, thus making the product out of use.

SUMMARY

[0004] Aiming at the problems in the prior art, the present disclosure provides a quick-detachable and foldable inflatable seat featuring simple structure, small size, regular folding, low risk of air leakage and easy disassembly and assembly.

[0005] Technical solutions of the present disclosure are as follows. A quick-detachable and foldable inflatable seat includes a first inflatable plate and a second inflatable plate. The first inflatable plate and the second inflatable plate, after being inflated, are assembled to form an inflatable seat by folding, the first inflatable plate being divided into a left support plate, a rear support plate and a right support plate, with creases arranged between the left support plate and the rear support plate and between the rear support plate and the right support plate, and the second inflatable plate being divided into a back plate and a seating plate, with a crease arranged therebetween, the first inflatable plate or upper and lower portions of the second inflatable plate being connected via sewing threads at the creases, and sealing strips being arranged on upper and lower sides of the sewing threads at the creases.

[0006] Further, a plurality of columns of sewing threads are arranged, and a plurality of layers of sealing strips are arranged.

[0007] Further, widths of the sealing strips in an inner layer is sequentially smaller than that of the sealing strips in an outer layer.

[0008] Further, two columns of sewing threads are arranged, and two layers of sealing strips are arranged.

[0009] Further, materials for making the first inflatable plate and the second inflatable plate include wire drawing cloth and sealing cloth, the sealing cloth being wrapped around the wire drawing cloth to connect upper and lower surfaces of the wire drawing cloth in a sealing manner to form hermetically sealed inflatable chambers.

[0010] Further, a first inflatable port is disposed on one of the left support plate, the rear support plate and the right support plate of the first inflatable plate.

[0011] Further, a second inflatable port is disposed on the back plate or the seating plate of the second inflatable plate.

[0012] Further, webbing hanging buckles are arranged on the left support plate and the right support plate of the first inflatable plate, and the webbing hanging buckles of the left support plate and the right support plate are fixedly connected via a webbing when folding and assembling are needed.

[0013] A method for treating creases of an inflatable plate includes the steps of: sewing two columns of high-strength sewing threads at creases by using a sewing machine, parallel spacing between the two columns of sewing threads being 1-3 cm and stitch spacing being 3-5 mm; reinforcing start points and end points of the sewing threads by back-stitching of 1-2 cm; and covering the sewing threads by thermally melting multi-layer sealing strips on upper and lower sides of the sewing threads by a hot-melt process.

[0014] Further, parallel spacing of the two columns of sewing threads is 2 cm and stitch spacing is 4 mm.

[0015] The present disclosure has the following advantageous effects.

[0016] 1. An existing multi-chambered integral spliced structure is changed thoroughly, and divided into two inflatable plates, and each inflatable plate can be inflated individually, which is convenient to check the leakage point.

[0017] 2. When folded, a single inflatable plate can be deflated for storage, and when stored, it can be folded into a regular shape without irregular folding shapes, preventing raw materials from damage and greatly reducing the risk of air leakage.

[0018] 3. For easy folding of an air cushion, the crease is treated by sewing using high-strength sewing threads, and the start points and end points of the sewing threads are reinforced by back-stitching, to prevent being burst by the air pressure when inflation is performed, and to ensure that the air cushion can be folded into a designed shape perfectly. In order to further reduce the risk of air leakage, the hot-melt process is used to thermally melt the multi-layer sealing strips on the upper and lower sides of the sewing threads to cover the sewing threads, not only preventing stitches from air leakage when inflation is performed, but also preventing the inflatable plates from damage caused by repeatedly folding.

[0019] 4. The two inflatable plates can be assembled into a finished inflatable seat after being inflated and folded, and can also be unfolded as a mattress during camping to insulate heat and resist damp.

[0020] 5. Webbing is used to connect the webbing hanging buckles for assembling, which is quick and convenient in assembly and disassembly.

[0021] 6. The seat can be used as a rescue plate after being disassembled in case of emergency rescues on water surface.

BRIEF DESCRIPTION OF THE DRAWINGS

[0022] FIG. 1 is an overall structural schematic diagram of an assembled state according to the present disclosure;

[0023] FIG. 2 is a schematic diagram of FIG. 1 from another perspective;

[0024] FIG. 3 is a front view of FIG. 1;

[0025] FIG. 4 is a right view of FIG. 1;

[0026] FIG. 5 is a schematic diagram of a first inflatable plate in an unfolding state;

[0027] FIG. 6 is a planar schematic diagram of the first inflatable plate;

[0028] FIG. 7 is a sectional view of the first inflatable plate;

[0029] FIG. 8 is a schematic diagram of the first inflatable plate in a folding state;

[0030] FIG. 9 is a top view of FIG. 8;

[0031] FIG. 10 is a planar schematic diagram of a second inflatable plate;

[0032] FIG. 11 is a sectional view of the second inflatable plate; and

[0033] FIG. 12 is a schematic diagram of the first inflatable plate in a folding state.

REFERENCE NUMERALS AND DENOTATIONS THEREOF

[0034] 1—first inflatable plate; 101—wire drawing cloth; 102—sealing cloth; 11—left support plate; 12—rear support plate; 13—right support plate; 14—first inflatable port; 2—second inflatable plate; 21—back plate; 22—seating plate; 23—second inflatable port; 3—sealing strip; 4—webbing hanging buckle; and 5—webbing.

DETAILED DESCRIPTION

[0035] The technical solutions of the examples in the present disclosure will be described clearly and completely by reference to the accompanying drawings of the examples in the present disclosure below. Obviously, the examples described are only some, rather than all examples of the present disclosure. Based on the examples of the present disclosure, all the other examples obtained by those ordinary skilled in the art without creative efforts fall within the scope of protection of the present disclosure.

[0036] As shown in FIGS. 1-12, the present disclosure provides a quick-detachable and foldable inflatable seat, including a first inflatable plate 1 and a second inflatable plate 2. The first inflatable plate 1 and the second inflatable plate 2, after being inflated, are assembled to form an inflatable seat by folding.

[0037] Materials for making the first inflatable plate 1 include wire drawing cloth 101 and sealing cloth 102. The sealing cloth 102 is wrapped around the wire drawing cloth 101 to connect upper and lower surfaces of the wire drawing cloth 101 in a sealing manner, so that a hermetically sealed inflatable chamber is formed between the sealing cloth 102 and the upper and lower surfaces of the wire drawing cloth 101. The sealing cloth 102 is polyvinyl chloride (pvc) cloth with a thickness of 0.5-1 mm.

[0038] The first inflatable plate 1 is divided into a left support plate 11, a rear support plate 12 and a right support plate 13, with creases arranged between the left support plate 11 and the rear support plate 12 and between the rear support plate 12 and the right support plate 13. Sewing threads are used to connect the upper and lower surfaces of the wire drawing cloth 101. Sealing strips 3 are arranged on upper and lower surfaces of the sowing threads at the crease. A plurality of columns of sewing threads are arranged, and a plurality of layers of sealing strips 3 are arranged, the sealing strip 3 being the pvc cloth with the thickness of 0.5-1 mm.

[0039] A first inflatable port 14 is disposed on one of the left support plate 11, the rear support plate 12 and the right support plate 13 of the first inflatable plate 1, and the left support plate 11, the rear support plate 12 and the right

support plate 13 of the first inflatable plate 1 are inflated and deflated through the first inflatable port 14.

[0040] Similarly, materials for making the second inflatable plate 2 include the wire drawing cloth 101 and the sealing cloth 102. The sealing cloth 102 is wrapped around the wire drawing cloth 101 to connect the upper and lower surfaces of the wire drawing cloth 101 in a sealing manner, so that a hermetically sealed inflatable chamber is formed between the sealing cloth 102 and the upper and lower surfaces of the wire drawing cloth 101.

[0041] The second inflatable plate 2 is divided into a back plate 21 and a seating plate 22, with a crease arranged therebetween. Sewing threads are used to connect upper and lower surfaces of the wire drawing cloth 101. Sealing strips 3 are arranged on upper and lower surfaces of the sowing threads at the crease. A plurality of columns of sewing threads are arranged, and a plurality of layers of sealing strips 3 are arranged.

[0042] A second inflatable port 23 is disposed on the back plate 21 or the seating plate 22 of the second inflatable plate 2, and the back plate 21 and the seating plate 22 of the second inflatable plate 2 are inflated and deflated through the second inflatable port 23.

[0043] Webbing hanging buckles 4 are arranged on the left support plate 11 and the right support plate 13 of the first inflatable plate 1. When folded and assembled, the left support plate 11 and the right support plate 13 are parallel and are perpendicularly distributed to the rear support plate 12, and the webbing hanging buckles 4 of the left support plate 11 and the right support plate 13 are fixedly connected by a webbing. The back plate 21 and the seating plate 22 of the second inflatable plate 2 are folded into an L-shape. The second inflatable plate 2 folded into the L-shape is placed between the folded first inflatable plate 1. The seating plate 22 of the second inflatable plate 2 is placed on a fixing net formed by the webbing 5, and the back plate 21 of the second inflatable plate 2 is fitted to an upper end portion of the rear support plate 12.

[0044] A method for treating creases of an inflatable plate includes the following steps. Firstly, two columns of high-strength sewing threads are sewed at creases by using a sewing machine, parallel spacing between the two columns of sewing threads being 1-3 cm and stitch spacing being 3-5 mm. As an optimal solution, parallel spacing of the two columns of sewing threads is 2 cm and stitch spacing is 4 mm. This design facilitates the ventilation between air chambers of the left support plate 11, the rear support plate 12 and the right support plate 13, and facilitates folding at the same time.

[0045] Secondly, start points and end points of the sewing threads are reinforced by back-stitching of 1-2 cm to prevent being burst by air pressure when inflation is performed, and the above process ensures that the seat can be folded into a designed shape perfectly.

[0046] Finally, in order to further reduce the risk of air leakage, a hot-melt process is used to thermally melt sealing strips 3 with widths of 4 cm on upper and lower sides of the sewing threads to cover the sewing threads, preventing stitches from air leakage when inflation is performed. In order to prevent the damage caused by repeatedly folding, a layer of sealing strips 3 with widths of 6 cm is pasted on the sealing strips 3 with widths of 4 cm, thus ensuring that the damage and air leakage caused by repeatedly folding can be

avoided. It has been proved by an experiment that the number of times of folding reaches more than 2600.

[0047] The above are only specific examples of the present disclosure, and cannot be used to limit the scope of implementation of the present disclosure, and therefore, the replacement of equivalent components, or equivalent changes and modifications made in accordance with the scope of patent protection of the present disclosure, still fall within the scope defined by the claims of the present disclosure.

1. A quick-detachable and foldable inflatable seat, comprising a first inflatable plate and a second inflatable plate, wherein the first inflatable plate and the second inflatable plate, after being inflated, are assembled to form an inflatable seat by folding, the first inflatable plate being divided into a left support plate, a rear support plate and a right support plate, with creases arranged between the left support plate and the rear support plate and between the rear support plate and the right support plate, and the second inflatable plate being divided into a back plate and a seating plate, with a crease arranged therebetween, the first inflatable plate or upper and lower portions of the second inflatable plate being connected via sewing threads at the creases, and sealing strips being arranged on upper and lower sides of the sewing threads at the creases.

2. The quick-detachable and foldable inflatable seat according to claim 1, wherein a plurality of columns of sewing threads are arranged, and a plurality of layers of sealing strips are arranged.

3. The quick-detachable and foldable inflatable seat according to claim 2, wherein widths of the sealing strips in an inner layer is sequentially smaller than that of the sealing strips in an outer layer.

4. The quick-detachable and foldable inflatable seat according to claim 3, wherein two columns of sewing threads are arranged, and two layers of sealing strips are arranged.

5. The quick-detachable and foldable inflatable seat according to claim 1, wherein materials for making the first inflatable plate and the second inflatable plate comprise wire drawing cloth and sealing cloth, the sealing cloth being wrapped around the wire drawing cloth to connect upper and lower surfaces of the wire drawing cloth in a sealing manner to form hermetically sealed inflatable chambers.

6. The quick-detachable and foldable inflatable seat according to claim 1, wherein a first inflatable port is disposed on one of the left support plate, the rear support plate and the right support plate of the first inflatable plate.

7. The quick-detachable and foldable inflatable seat according to claim 1, wherein a second inflatable port is disposed on the back plate or the seating plate of the second inflatable plate.

8. The quick-detachable and foldable inflatable seat according to claim 1, wherein webbing hanging buckles are arranged on the left support plate and the right support plate of the first inflatable plate, and the webbing hanging buckles of the left support plate and the right support plate are fixedly connected via a webbing when folding and assembling are needed.

9. A method for treating creases of an inflatable plate, comprising the steps of: sewing two columns of high-strength sewing threads at creases by using a sewing machine, parallel spacing between the two columns of sewing threads being 1-3 cm and stitch spacing being 3-5 mm; reinforcing start points and end points of the sewing threads by back-stitching of 1-2 cm; and covering the sewing threads by thermally melting multi-layer sealing strips on upper and lower sides of the sewing threads by a hot-melt process.

10. The method for treating creases of an inflatable plate according to claim 9, wherein parallel spacing of the two columns of sewing threads is 2 cm and stitch spacing is 4 mm.

* * * * *